

Lab 52 Worksheet — Backup Strategy Design and the 3-2-1 Rule

Name: ____ Date: ____

Exam objective: Design a backup strategy specifying types, rotation, retention and off-site copy, and justify it against a stated recovery objective (Core 2 objective 4.3).

Record as you go

Step	What you did	What you observed
1	Record the two objectives that drive every backup design: the recovery point objective, which is how much data the business can afford to lose, and the recovery time objective, which is how long it can afford to be down.	
2	Record the scenario objectives: a small business with a 24-hour RPO and a 4-hour RTO for its file server.	
3	Record the 3-2-1 rule and what each element defeats: three copies defeat corruption, two different media defeat media failure, and one off-site copy defeats fire, flood and theft.	
4	Compare full, incremental and differential backups on backup time, storage consumed, restore complexity and archive bit behaviour.	
5	Work the restore arithmetic for an incremental scheme: a Sunday full plus daily incrementals means restoring Thursday requires the full plus Monday, Tuesday, Wednesday and Thursday — five operations.	
6	Work the same arithmetic for a differential scheme: a Sunday full plus daily differentials means restoring Thursday requires only the full plus Thursday — two operations.	

Step	What you did	What you observed
7	Choose between them against the 4-hour RTO, and justify the choice on restore time rather than on backup window alone.	
8	Record the rotation schemes: grandfather-father-son with daily, weekly and monthly sets, and the tower of Hanoi scheme, and state what each provides.	
9	Define the retention policy: how long daily, weekly, monthly and yearly copies are kept, and note that retention may be set by regulation rather than by preference.	
10	Specify the off-site copy: where it goes, how it gets there, how often, and whether it is encrypted in transit and at rest.	
11	Record the ransomware requirement: at least one copy must be offline, air-gapped or immutable, because ransomware encrypts every backup it can reach over the network.	
12	Define the backup testing schedule, specifying how often a restore is actually performed and who signs off that it succeeded.	

Verification

Success criterion: Your strategy states both RPO and RTO and the schedule meets them; the restore arithmetic is worked for both incremental and differential; the choice is justified against the 4-hour RTO; and the design includes an offline or immutable copy and a stated restore test schedule.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
